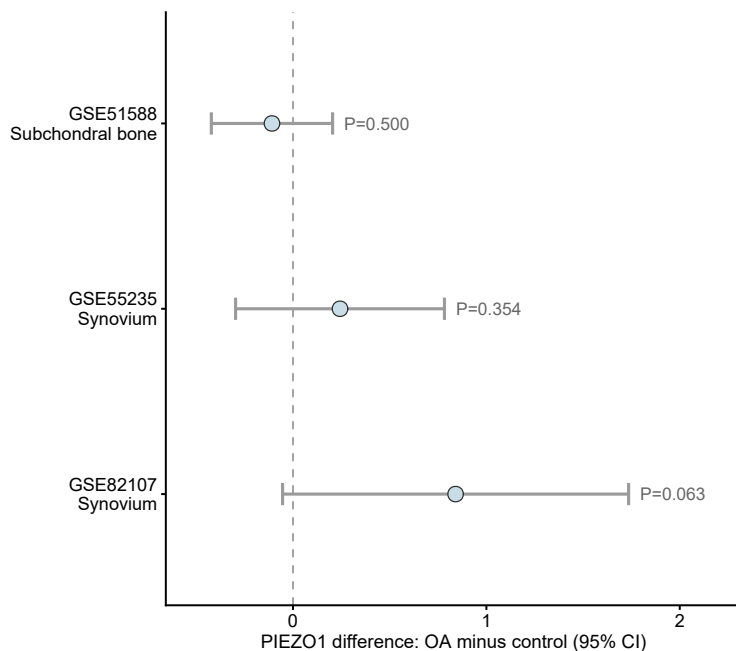


a

Evidence layer	Data and sample structure	Inferential unit
Disease-level expression	Four tissue datasets	Sample within dataset
Paired joint compartments	GSE51588: 20 OA + 5 non-OA donors	Paired donor difference
Donor-aware states	GSE104782: 1,464 cells; 10 donors	One estimate per donor
Cross-dataset replication	GSE57218: 33 pairs; GSE152805: 3 pairs	Paired donor difference
Within-state decomposition	GSE152805: 7 states x 3 donors	Donor x state mean
Blood-expression genetics	eQTLGen cis-eQTL x FinnGen R12	Instrument-level Wald ratio

b

Disease-level contrasts are inconsistent



c

Evidence layer	SUPPORTS	DOES NOT SUPPORT
Paired joint compartments	Compartment-associated regulation within OA joints, free of between-person confounding	A direct measurement of mechanical force, or proof that the contrast is OA specific
Donor-aware chondrocyte states	Association of PIEZO1 with the state continuum at the level of the donor	Cell-level independent replication, or that the state transition is causal
Programme-score replication	A reproducible programme-level contrast across three paired datasets	Inferred cell-type abundance: similarity weights are not proportions
Within-state decomposition	Separation of within-state change from compositional shift	Statistical replication: three donors support direction only
Cross-tissue signature transfer	Directional consistency of regional programmes across tissues	Equivalence of PIEZO1 direction between bone and cartilage
Blood-expression Mendelian randomization	Upper bounds on a systemic effect mediated by blood expression	Any local joint-tissue mechanism, or tissue-specific regulation